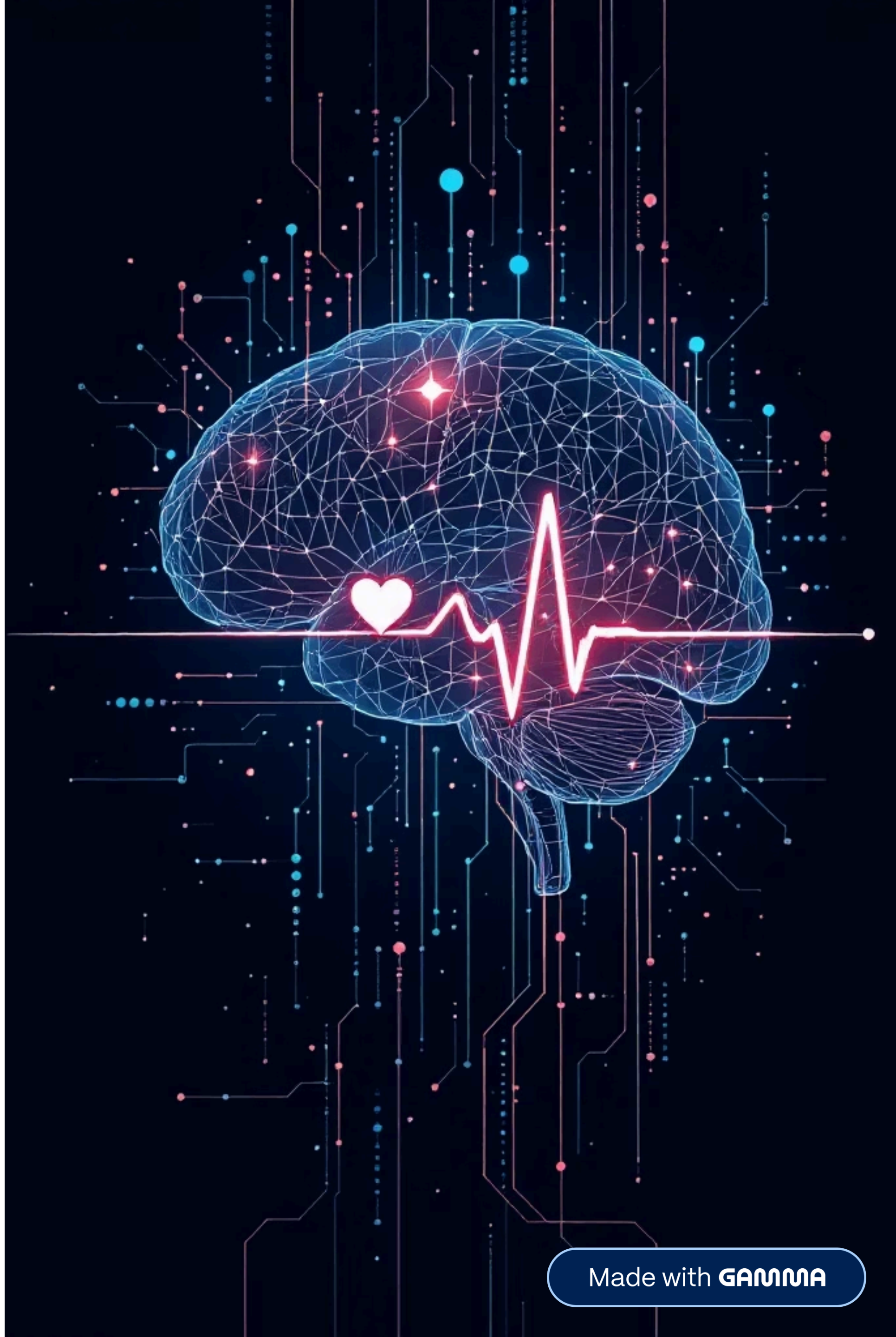
🧠 Talk2Mind

Multimodal Mental Well-Being Assessment using Deep Learning

Bhautik Gondaliya

AI | Deep Learning | Multimodal Emotion Recognition

Links: [GitHub] and [Demo Video]



Project Overview

Problem Statement

Mental health monitoring is critical but limited by subjective assessments

Why It Matters

Early detection and continuous monitoring improve outcomes

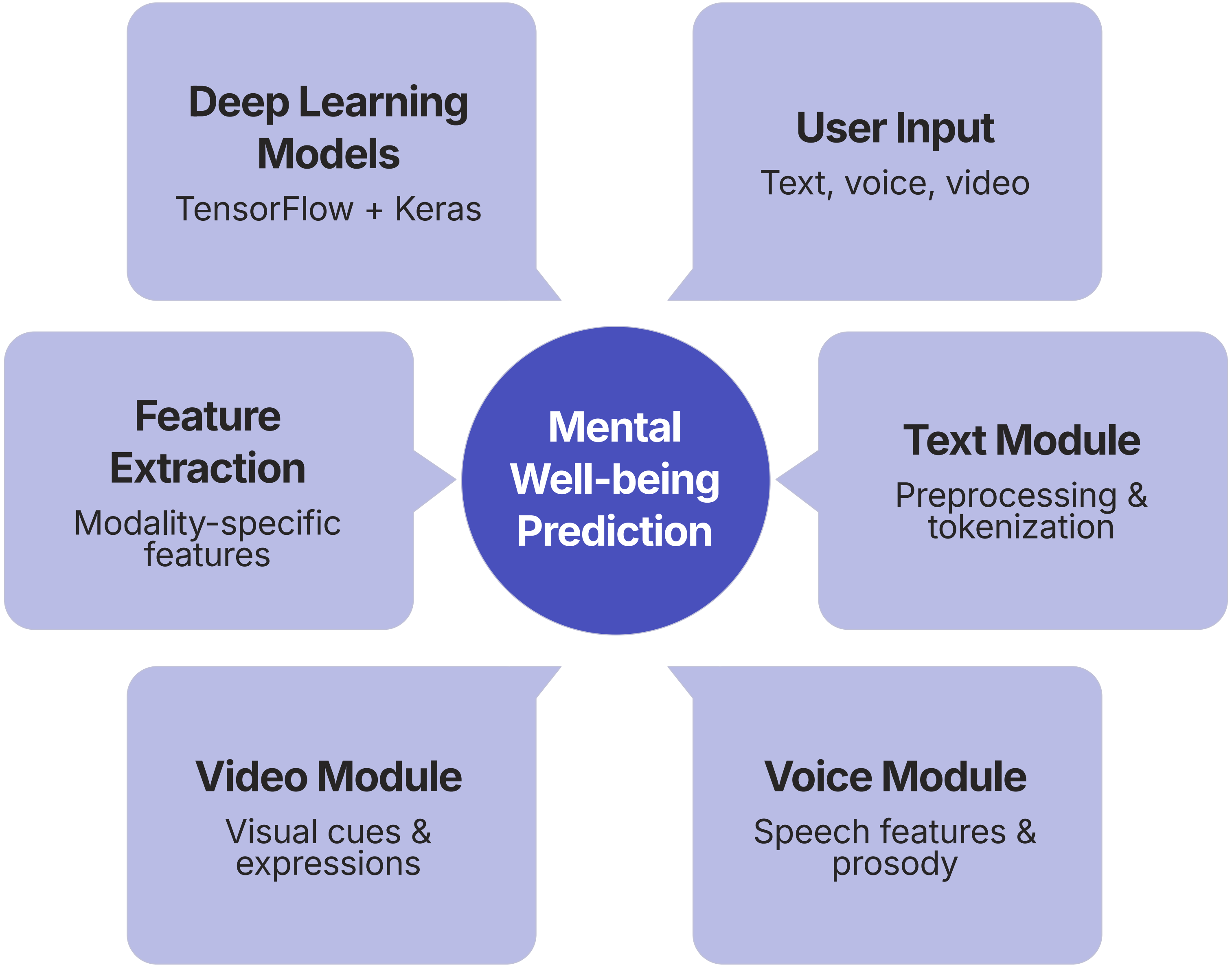
Existing Limitations

Current methods lack multimodal analysis and real-time insights

Our Solution

Talk2Mind: AI-powered multimodal assessment for holistic mental wellness

System Architecture



AI Models

Text Emotion Recognition

- Embedding
- BiLSTM
- Dense Layers
- Emotion Prediction

Speech Emotion Recognition

- MFCC Features
- Mel Spectrogram
- CNN
- LSTM

Facial Emotion Recognition

- Face Detection
- CNN Backbone
- LSTM
- Emotion Prediction

Late Fusion Model

- Feature Concatenation
- Dense Layers
- Mental Wellness Score

Dataset & Technologies

Datasets

- CMU-MOSEI
- TESS
- RAVDESS

[1] Technology Stack

- Python
- TensorFlow & Keras
- OpenCV
- Librosa
- FastAPI
- Streamlit
- Google Colab
- GitHub

Preprocessing

Training

Inference

Deployment

Application Demo & Results

User

Text Check-in
(What's on your mind?)

Voice Check-in
(How are you feeling?)

Face Check-in
(Facial expression analysis)

Dashboard & Results

Emotion Detection
Real-time emotion analysis

Mental Wellness Score
Comprehensive wellness metrics

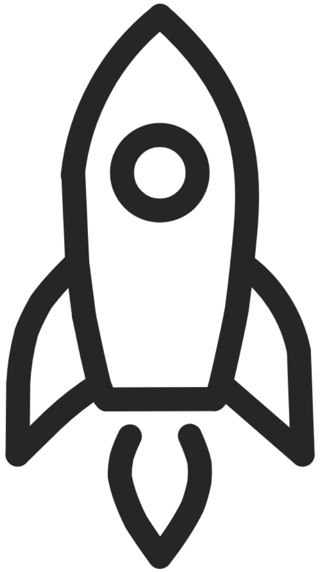
Risk Assessment
Early warning indicators

Recommendations
Personalized suggestions

Conclusion & Future Scope

Achievements

- Multimodal AI Integration
- Deep Learning Models
- Healthcare AI Application
- Interactive Dashboard
- Real-time Inference



Future Scope

- Live Camera Detection
- Mobile Application
- Therapist Integration
- Wearable Device Support
- Continuous Monitoring

Thank You

Questions?

GitHub | Demo Video